Water Index Quality(WQI) can be reliably predicted using a stacking model with fewer, low-cost parameters than the traditional WQI formula

Title: Machine Learning-based Water Quality Index Prediction in the Quang Ninh Province, Vietnam

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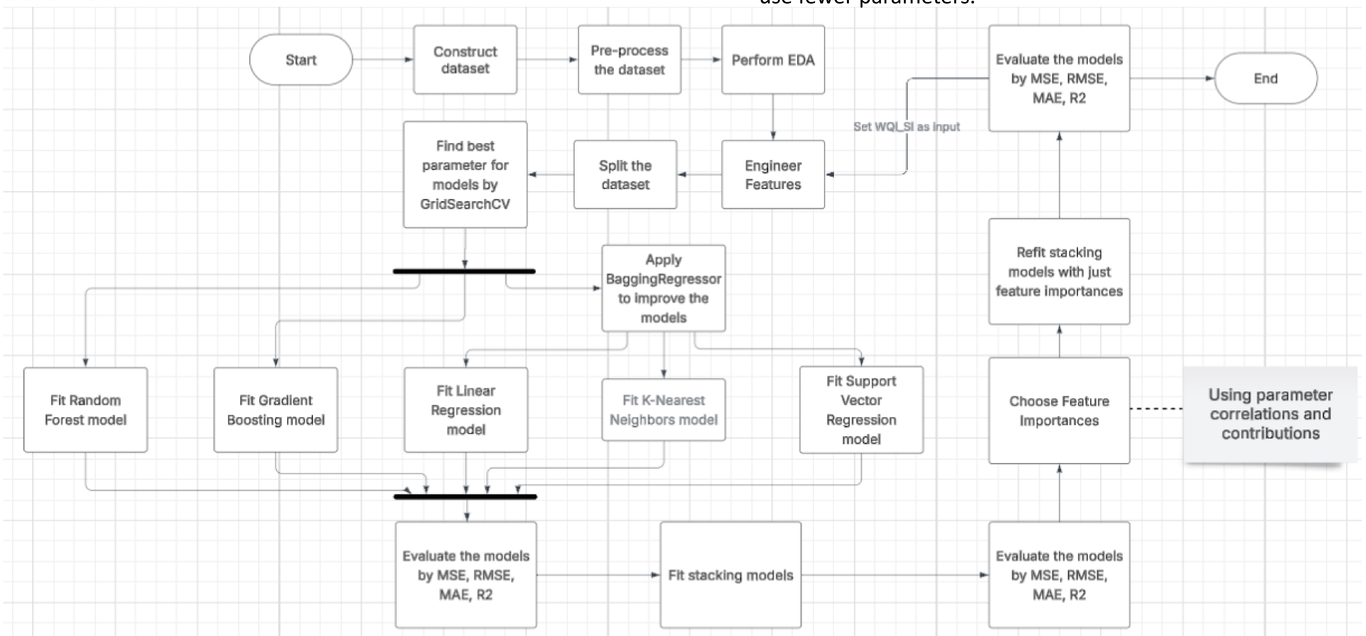
OBJECTIVE:

- Execute feature selection
- To optimize model performance through a data-driven and data-centric AI approach

BACKGROUND:

- Water quality management is critical, especially in Quang Ninh Province and Ha Long Bay (UNESCO site).
- Traditional WQI calculation is costly, time-consuming, and risky due to extensive field sampling, while WQI calculated with fewer parameters is less accurate.
- AI-based WQI prediction is a recent trend due to its ability to use fewer parameters.

WORKFLOW



SUMMARY OF RESULTS

- Removed As, Cd, NH_4,NO_3, BOD_5, Coliform
- Under reduced input conditions, the machine learning model demonstrated superior predictive performance relative to the conventional WQI calculation.

Model	MSE	RMSE	MAE	R^2
Best Stacking model	3.67	1.92	0.98	0.97
Traditional WQI calculation	199.9	14.14	13.97	-0.56

DICUSSION

- Instead of using raw parameters, WQI sub-indices were used as model inputs, resulting in better model performance.
- While previous studies often report that tree-based models perform best due to their robustness to outliers, our results show that after applying the transformed inputs, Linear Regression and Support Vector Regression achieved better performance.
- In practice, standardization performed better than robust standardization, which is consistent with findings reported in some existing studies.
- Using permutation importance for feature selection may remove critical information, making it harder to predict minority cases.

DETAIL RESULTS

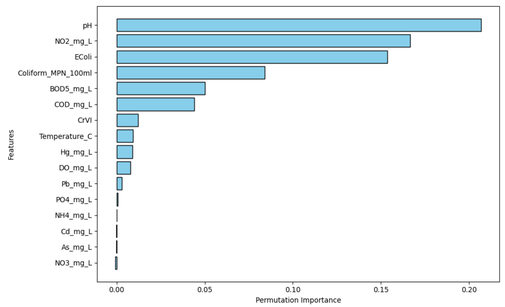
Direct Measurement Inputs

Before removing unimportant features

Model	MSE	RMSE	MAE	R^2
Random Forest	4.97	2.23	1.06	0.96
Gradient Boosting	4.04	2.01	0.91	0.97
Linear Regression	27.06	5.20	3.51	0.77
K-Nearest Neighbors	14.34	3.79	2.15	0.88
Support Vector Regression	10.31	3.21	1.75	0.91

Stacking Models Performance Metrics

Base Models	Num Models	MSE	RMSE	MAE	R2
KNN + SVR + GradientBoosting	3	3.777	1.9434	0.8738	0.9678
SVR + GradientBoosting	2	3.7792	1.944	0.8746	0.9678
LinearRegression + SVR + GradientBoosting	3	3.8292	1.9568	0.869	0.9673
KNN + LinearRegression + SVR + GradientBoosting	4	3.8384	1.9592	0.8662	0.9673
RandomForest + KNN + SVR + GradientBoosting	4	3.9324	1.983	0.8855	0.9665
RandomForest + SVR + GradientBoosting	3	3.9324	1.983	0.8855	0.9665
RandomForest + LinearRegression + SVR + GradientBoosting	4	3.9588	1.9897	0.8822	0.9662
RandomForest + KNN + LinearRegression + SVR + GradientBoosting	5	3.9681	1.992	0.8807	0.9662
RandomForest + SVR	2	4.1655	2.041	1.102	0.9645
RandomForest + KNN + SVR	3	4.2377	2.0586	1.0922	0.9639
RandomForest + LinearRegression + SVR	3	4.3218	2.0789	1.1021	0.9631
RandomForest + KNN + LinearRegression + SVR	4	4.3242	2.0795	1.0979	0.9631
KNN + GradientBoosting	2	4.3499	2.0856	0.9191	0.9629
KNN + LinearRegression + GradientBoosting	3	4.4022	2.0981	0.9139	0.9625
LinearRegression + GradientBoosting	2	4.4649	2.113	0.92	0.9619
RandomForest + KNN + GradientBoosting	3	4.5198	2.126	0.9379	0.9615
RandomForest + KNN + LinearRegression + GradientBoosting	4	4.5469	2.1324	0.9317	0.9612
RandomForest + GradientBoosting	2	4.6107	2.1473	0.9524	0.9607
RandomForest + LinearRegression + GradientBoosting	3	4.6214	2.1497	0.9382	0.9606
RandomForest + KNN	2	5.1454	2.2683	1.1274	0.9561
RandomForest + KNN + LinearRegression	3	5.1901	2.2782	1.133	0.9557
RandomForest + LinearRegression	2	5.3446	2.3118	1.1681	0.9544
KNN + LinearRegression + SVR	3	7.0748	2.6598	1.5967	0.9397
KNN + SVR	2	7.3172	2.705	1.6531	0.9376
LinearRegression + SVR	2	7.9862	2.826	1.75	0.9319
KNN + LinearRegression	2	12.2143	3.4949	2.1594	0.8958



After removing unimportant features

Stacking Models Performance Metrics

Base Models	Num Models	MSE	RMSE	MAE	R2
RandomForest + LinearRegression + SVR + GradientBoosting	4	5.0658	2.2507	1.1276	0.9568
RandomForest + SVR + GradientBoosting	3	5.0711	2.2519	1.1278	0.9568
LinearRegression + SVR + GradientBoosting	3	5.0923	2.2566	1.1356	0.9566
SVR + GradientBoosting	2	5.0967	2.2576	1.1358	0.9565
RandomForest + KNN + SVR + GradientBoosting	4	5.1136	2.2613	1.141	0.9564
KNN + SVR + GradientBoosting	3	5.1307	2.2651	1.1467	0.9563
RandomForest + KNN + LinearRegression + SVR + GradientBoosting	5	5.1652	2.2727	1.147	0.956
KNN + LinearRegression + SVR + GradientBoosting	4	5.1817	2.2763	1.1519	0.9558
RandomForest + LinearRegression + SVR	3	5.5295	2.3515	1.2094	0.9529
RandomForest + SVR	2	5.5513	2.3561	1.2103	0.9527
RandomForest + KNN + LinearRegression + SVR	4	5.6436	2.3756	1.2197	0.9519
RandomForest + KNN + GradientBoosting	3	5.7039	2.3883	1.2176	0.9514
KNN + GradientBoosting	2	5.7199	2.3916	1.2225	0.9512
RandomForest + KNN + SVR	3	5.7296	2.3937	1.2275	0.9511
RandomForest + KNN + LinearRegression + GradientBoosting	4	5.7739	2.4029	1.2253	0.9508
KNN + LinearRegression + GradientBoosting	3	5.7835	2.4049	1.2275	0.9507
RandomForest + GradientBoosting	2	5.8648	2.4217	1.2196	0.95
RandomForest + LinearRegression + GradientBoosting	3	5.893	2.4275	1.2221	0.9498
LinearRegression + GradientBoosting	2	5.9377	2.4367	1.2305	0.9494
RandomForest + KNN	2	6.9626	2.6387	1.3148	0.9406
RandomForest + KNN + LinearRegression	3	6.9788	2.6417	1.318	0.9405
RandomForest + LinearRegression	2	6.9918	2.6442	1.2932	0.9404
LinearRegression + SVR	2	10.0087	3.1636	1.896	0.9147
KNN + LinearRegression + SVR	3	10.3115	3.2112	1.9269	0.9121
KNN + SVR	2	10.9703	3.3121	1.9899	0.9065
KNN + LinearRegression	2	20.5557	4.5338	2.8459	0.8247

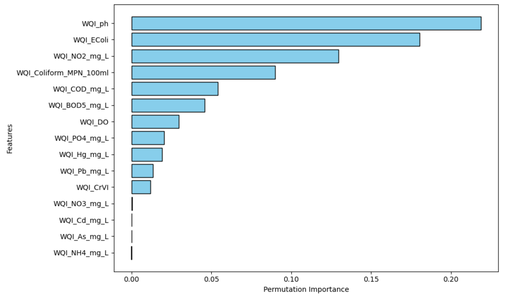
Feature Engineered Inputs (WQI_SI)

Before removing unimportant features

Model	MSE	RMSE	MAE	R^2
Random Forest	3.54	1.88	0.71	0.97
Gradient Boosting	3.43	1.85	0.75	0.97
Linear Regression	0.75	0.87	0.45	0.99
K-Nearest Neighbors	2.84	1.69	0.70	0.98
Support Vector Regression	0.73	0.85	0.40	0.99

Stacking Models Performance Metrics

Base Models	Num Models	MSE	RMSE	MAE	R2
LinearRegression + SVR	2	0.3035	0.5509	0.3548	0.9974
KNN + LinearRegression + SVR	3	0.3336	0.5776	0.3563	0.9972
RandomForest + LinearRegression + SVR	3	0.3437	0.5862	0.3575	0.9971
RandomForest + KNN + LinearRegression + SVR	4	0.3527	0.5939	0.3593	0.997
KNN + LinearRegression + SVR + GradientBoosting	4	0.4323	0.6575	0.3729	0.9963
LinearRegression + SVR + GradientBoosting	3	0.4359	0.6602	0.3724	0.9963
RandomForest + KNN + LinearRegression + SVR + GradientBoosting	5	0.4407	0.6638	0.3814	0.9962
RandomForest + LinearRegression + SVR + GradientBoosting	4	0.4427	0.6654	0.3805	0.9962
KNN + LinearRegression	2	0.5464	0.7392	0.4057	0.9953
RandomForest + KNN + LinearRegression	3	0.5845	0.7645	0.4036	0.995
KNN + LinearRegression + GradientBoosting	3	0.6836	0.8268	0.4204	0.9942
RandomForest + KNN + LinearRegression + GradientBoosting	4	0.6897	0.8305	0.4284	0.9941
RandomForest + SVR	2	0.7514	0.8668	0.4734	0.9936
RandomForest + KNN + SVR	3	0.7653	0.8748	0.475	0.9935
RandomForest + LinearRegression	2	0.7975	0.893	0.4218	0.9932
KNN + SVR + GradientBoosting	3	0.8148	0.9027	0.4541	0.9931
SVR + GradientBoosting	2	0.8186	0.9048	0.4537	0.993
RandomForest + KNN + SVR + GradientBoosting	4	0.8326	0.9125	0.4686	0.9929
RandomForest + SVR + GradientBoosting	3	0.838	0.9154	0.4637	0.9929
KNN + SVR	2	0.9361	0.9675	0.4573	0.992
RandomForest + LinearRegression + GradientBoosting	3	0.9629	0.9813	0.4579	0.9918
LinearRegression + GradientBoosting	2	0.9901	0.9951	0.4696	0.9916
KNN + GradientBoosting	2	1.9233	1.3868	0.6348	0.9836
RandomForest + KNN + GradientBoosting	3	1.9273	1.3883	0.6469	0.9836
RandomForest + KNN	2	2.18	1.4765	0.7225	0.9814
RandomForest + GradientBoosting	2	3.4539	1.8585	0.7392	0.9705



After removing unimportant features

Stacking Models Performance Metrics

Base Models	Num Models	MSE	RMSE	MAE	R2
SVR + GradientBoosting	2	3.666	1.9147	0.9765	0.9687
KNN + SVR + GradientBoosting	3	3.6672	1.915	0.9769	0.9687
RandomForest + KNN + SVR + GradientBoosting	4	3.675	1.917	0.9772	0.9687
RandomForest + SVR + GradientBoosting	3	3.6794	1.9182	0.9787	0.9686
RandomForest + KNN + LinearRegression + SVR + GradientBoosting	5	3.7513	1.9366	0.9882	0.968
KNN + LinearRegression + SVR + GradientBoosting	4	3.759	1.9388	0.9908	0.9679
RandomForest + LinearRegression + SVR + GradientBoosting	4	3.7746	1.9441	0.9962	0.9678
LinearRegression + SVR + GradientBoosting	3	3.7814	1.9446	0.9964	0.9678
RandomForest + KNN + LinearRegression + GradientBoosting	4	3.8823	1.9703	1.0088	0.9669
KNN + LinearRegression + GradientBoosting	3	3.8879	1.9718	1.0096	0.9668
RandomForest + SVR	2	3.9232	1.9807	0.9945	0.9665
RandomForest + KNN + SVR	3	3.947	1.9867	0.9984	0.9663
RandomForest + KNN + LinearRegression + SVR	4	3.991	1.9977	1.0049	0.966
RandomForest + LinearRegression + SVR	3	4.0163	2.0041	1.008	0.9658
KNN + LinearRegression	2	4.0937	2.0233	0.99	0.9651
KNN + LinearRegression + SVR	3	4.098	2.0244	0.9741	0.9651
KNN + GradientBoosting	2	4.1387	2.0344	1.0227	0.9647
RandomForest + KNN + GradientBoosting	3	4.1449	2.0359	1.0237	0.9647
RandomForest + KNN + LinearRegression	3	4.1511	2.0374	1.0289	0.9646
LinearRegression + GradientBoosting	2	4.202	2.0499	1.0638	0.9642
RandomForest + LinearRegression + GradientBoosting	3	4.2372	2.0584	1.0617	0.9639
LinearRegression + SVR	2	4.2704	2.0665	0.9918	0.9636
KNN + SVR	2	4.4401	2.109	1.0302	0.9621
RandomForest + LinearRegression	2	4.5862	2.1415	1.0743	0.9609
RandomForest + KNN	2	4.8183	2.1951	1.1043	0.9589
RandomForest + GradientBoosting	2	5.4213	2.3284	1.1316	0.9538

Descriptive Statistics of Water Quality Parameters

Parameter	Unit	Minimum	Maximum	Mean	Median	Std. Deviation
Temperature	Degree Celsius	7.4	39.4	24.222	23.6	4.843
pH	pH unit	4.59	8.64	6.685	6.64	0.444
DO	mg/L	1.08	9.7	6.703	6.72	0.997
COD	mg/L	1.5	306.4	15.604	11.3	23.158
BOD_5	mg/L	1	54.4	4.28	3.1	4.192
NO_3	mg/L	0.008	6.528	0.345	0.214	0.638
NO_2	mg/L	0.002	8.337	0.185	0.002	0.951
NH_4	mg/L	0.01	6.321	0.399	0.054	1.018
PO_4	mg/L	0.01	10.365	0.051	0.01	0.336
Coliform	MPN/100ml	3	25000	766.685	150	1885.311
EColi		3	400	14.118	3	35.718
As	mg/L	0	0.014	0.001	0	0.001
Hg	mg/L	0	0.006	0	0	0
Pb	mg/L	0	0.066	0.001	0.001	0.003
Cd	mg/L	0	0.011	0.001	0.001	0
CrVI		0.001	0.03	0.001	0.001	0.003



